

Software Requirements Specification

Board Game Tutor — Interactive Board Game Rules Learning Platform

Basic Software Development · Analysis and Requirements

Team name	PhomPen Duke
Project title	Board Game Tutor
SDLC chosen in M1 + reason	We chose the Agile software development model because this system relies on consistent testing based on feedback statements, as our goal is to implement a proven and easy-to-understand learning method for the board game. We will develop the system in rounds (iterations) and continuously evaluate whether each round understands the rules.
Date / version	9 October 2026 · version 1.1
GitHub repo	NONE

1. Introduction

1.1 Purpose and scope

New and casual board game players often struggle to understand the rules of a new game. Traditional rulebooks can be long and complex, while tutorial videos may require players to pause, replay, or search for specific information. As a result, players may spend a significant amount of time learning the rules before they can start playing.

The Board Game Learning Platform is an interactive learning system designed to help users understand board game rules in a more structured and practical way. Instead of only presenting a complete rulebook, the system guides users through the game step by step and allows them to practice with example game situations.

The system includes an AI Game Rule Assistant. Users can ask questions about the selected game, receive explanations grounded in relevant information retrieved from that game's Knowledge Base, and ask follow-up questions using the context of the ongoing conversation. The assistant supports learning and clarification; it does not replace official rules or content review.

The MVP is deliberately limited to a small number of supported board games. Its main purpose is to help users learn essential rules, understand how those rules are applied, and quickly find specific rule information when needed.

Success looks like this: a new player can learn the essential rules of a supported board game and feel prepared to start playing without repeatedly searching through a long rulebook or multiple tutorial sources.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
New / Casual Player (primary user)	primary user	Learn the rules of a board game clearly and quickly	Being confused by complex or unclear instructions

Name / role	Kind	What they need	What they fear / forbid
Game Host	secondary user	Help a group understand the game before playing	Spending too much time explaining the same rules repeatedly
Board Game Content Administrator	system administrator	Manage accurate game information, rules, and learning content	Incorrect or outdated information being published

2. Overall description

2.1 Product context — what it does / does not

The Board Game Learning Platform helps users learn selected board games through interactive tutorials, practical situations, and rule assistance.

The system is not simply a collection of rulebooks. It focuses on helping users understand how rules work through a structured learning process.

Everything below defines the MVP for this project.

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
- FR-1 Search and select a board game. - FR-2 Learn through an interactive tutorial. - FR-3 Practice with example game situations. - FR-4 Check understanding of game rules. - FR-5 Ask the AI Game Rule Assistant about specific game rules. - FR-6 Track learning progress. - FR-7 Manage board game content. - FR-8 Ask follow-up questions with conversational context.	- Online multiplayer board game gameplay. - Buying or selling board games. - Payment or subscription features. - Physical board game rental. - Delivery services. - Full digital recreation of board games. - Supporting all board games in the market.

2.2 Assumptions and constraints

Assumptions

- Users have access to a smartphone, tablet, or computer with an Internet connection.
- Users may have little or no experience with a selected board game.
- Users are willing to interact with learning content before starting a game.
- Board game information used in the system will be reviewed by the team before publication.

Constraints

- Constraint: The MVP will support only a limited number of board games.
- Constraint: The system will focus on learning and rule assistance rather than providing complete digital gameplay.
- Constraint: The project must be completed within the available time and resources of the development team.
- Constraint: Game rules and learning content must be checked for accuracy before being published.

Technical design note: the proposed implementation uses retrieval-augmented generation (RAG) to retrieve relevant rules from a game-specific Knowledge Base and provide them as context to a language model. The functional requirements describe user-visible behavior; specific model and runtime choices are implementation details.

3. Functional requirements

Format: As a <role>, I want <goal>, so that <benefit>. Acceptance criteria are written so each one could actually be tested.

FR-1 — Search and Select a Board Game

User story:

As a player, I want to search for a board game, so that I can find the game I want to learn.

Acceptance criteria:

1. Given that a player enters a board game name, when they perform a search, then the system shall display matching board games.
2. Given that search results are displayed, when the player selects a board game, then the system shall display the available learning content for that game.

FR-2 — Interactive Board Game Tutorial

User story:

As a new player, I want to learn a board game step by step, so that I can understand the basic rules before playing.

Pain this traces to: Players find long rulebooks difficult to understand.

Acceptance criteria:

1. Given that a player selects a supported board game, when they start the tutorial, then the system shall display the learning steps in the correct sequence.
2. Given that the player completes a learning step, when they continue, then the system shall display the next step.
3. The system shall allow the player to review previous learning steps.

FR-3 — Practice Game Situations

User story:

As a player, I want to practice with example game situations, so that I can understand how the rules are applied during gameplay.

Pain this traces to: Players do not know how rules apply in real situations.

Acceptance criteria:

1. Given an example game situation, the system shall present the relevant game information to the player.
2. The player shall be able to select or provide an answer based on the situation.
3. After the player submits an answer, the system shall provide feedback and an explanation.

FR-4 — Rule Understanding Check

User story:

As a player, I want to check my understanding of the game rules, so that I can know whether I am ready to play.

Pain this traces to: Players are unsure whether they understand the rules.

Acceptance criteria:

1. Given that a player completes the required tutorial content, when they start an understanding check, then the system shall display questions related to the selected board game.
2. When the player submits an answer, the system shall indicate whether the answer is correct.
3. When an answer is incorrect, the system shall provide an explanation or reference to the relevant rule.

FR-5 — AI Game Rule Assistant**User story:**

As a player, I want to ask questions about a specific game rule during gameplay and receive an understandable explanation, so that I can resolve rule-related questions without reading the entire rulebook.

Pain this traces to: Players cannot quickly find a specific rule during gameplay.

Acceptance criteria:

1. The player shall be able to enter a question about the selected game's rules.
2. The system shall retrieve relevant information from that game's Knowledge Base.
3. The AI shall generate an understandable response using the retrieved information as context.
4. Relevant rule references or retrieved sources shall be displayed when available.
5. If sufficient supporting information cannot be found, the system shall state this limitation rather than presenting unsupported information as confirmed fact.
6. If the AI service is unavailable, the system shall display an understandable error message.

FR-6 — Learning Progress Tracking**User story:**

As a player, I want to track my learning progress, so that I know which tutorials I have completed and which parts remain unfinished.

Pain this traces to: Players forget where they stopped learning.

Acceptance criteria:

1. The system shall record completed learning steps for each user.
2. When a player returns to a previously started tutorial, the system shall display their current progress.
3. The player shall be able to view completed and incomplete learning content.

FR-7 — Manage Board Game Content**User story:**

As an administrator, I want to manage board game information and learning content, so that users can access accurate and updated content.

Pain this traces to: Learning content needs to be maintained.

Acceptance criteria:

1. An authorized administrator shall be able to add a new board game.
2. An authorized administrator shall be able to add or update tutorials and rules.
3. An authorized administrator shall be able to manage example game situations.
4. Changes to content shall be saved and reflected in the system.

FR-8 — Follow-up Questions and Conversational Context

User story:

As a player, I want to ask follow-up questions about an AI response without repeating my original question, so that I can clarify rules through a continuous conversation.

Pain this traces to: Players need to clarify an AI answer without repeating the full question.

Acceptance criteria:

1. Given that the player has received an AI response, when they ask a follow-up question, then the system shall accept it in the same conversation.
2. The system shall use relevant earlier messages to interpret the follow-up question.
3. The system shall retain the selected board game as context and retrieve relevant Knowledge Base information where appropriate.
4. The response shall be relevant to both the conversation and the retrieved information.
5. The player shall be able to continue asking follow-up questions in the same conversation.
6. If the follow-up question is ambiguous, the system shall ask for clarification; if supporting information is unavailable, the system shall state that limitation.

4. Non-functional requirements

ID	Kind	Statement (testable)	How we will check
NFR-1	Performance	The system shall display board game search results within 2 seconds under normal network conditions.	Perform multiple searches and measure the response time.
NFR-2	Usability	A first-time user shall be able to start a board game tutorial within 30 seconds without assistance.	Conduct a usability test with first-time users.
NFR-3	Reliability	The system shall save a user's learning progress so that it remains available after the user leaves and returns to the system.	Start a tutorial, complete several steps, leave the system, return, and verify the saved progress.
NFR-4	Security	Users shall only be able to access and modify their own account information and learning progress.	Attempt to access another user's data without authorization.
NFR-5	Content Accuracy	Board game learning content shall be reviewed by an authorized administrator before being published to users.	Verify that unpublished or unapproved content cannot be accessed by normal users.
NFR-6	AI Response Reliability	AI rule explanations shall be grounded in relevant information retrieved from the selected game's Knowledge Base when such information is available. Sources shall be shown when available, and insufficient evidence shall be communicated.	Test supported questions, questions not covered by the Knowledge Base, and source display.

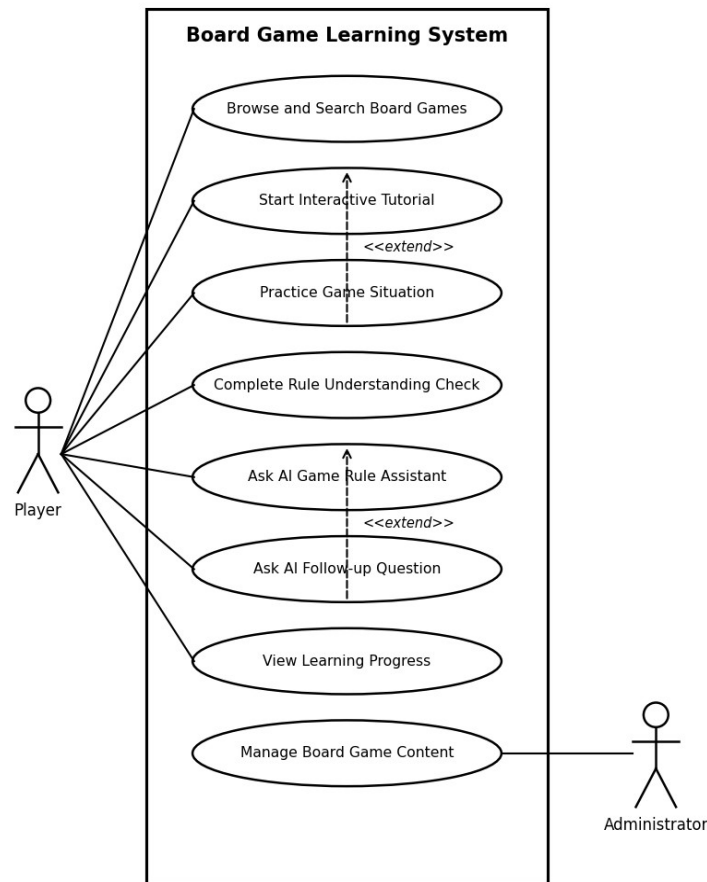
ID	Kind	Statement (testable)	How we will check
NFR-7	AI Response Time	The team shall measure and document AI response time in the intended deployment and demonstration environment. A target threshold shall be agreed after baseline testing because performance depends on hardware, model loading, and response length.	Run a defined set of test questions and record response times.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1	Player	FR-1	Search for a Board Game
UC-2	Player	FR-2	Start an Interactive Tutorial
UC-3	Player	FR-3	Practice a Game Situation
UC-4	Player	FR-4	Check Rule Understanding
UC-5	Player	FR-5	Ask the AI Game Rule Assistant
UC-6	Player	FR-6	View Learning Progress
UC-7	Administrator	FR-7	Manage Board Game Content
UC-8	Player	FR-8	Ask an AI Follow-up Question

5.2 Use-case diagram



5.3 UC-8 — Ask an AI Follow-up Question

Primary actor	Player.
Preconditions	The player has selected a supported board game and has an AI conversation open.
Main flow	1. The player asks a rule question. 2. The system retrieves relevant Knowledge Base information and responds. 3. The player enters a follow-up question referring to the earlier response. 4. The system uses relevant conversation context and retrieves supporting rules. 5. The system responds and displays sources when available.
Alternative flow	If the follow-up question is ambiguous, the system asks for clarification. If supporting information cannot be found, the system communicates that limitation rather than claiming an unsupported answer.

6. Out of scope

1. Online multiplayer board game gameplay.
2. Buying and selling board games.
3. Payment and subscription systems.

4. Physical board game rental services.
5. Delivery services.
6. Full digital versions of physical board games.
7. Supporting every board game available on the market.

7. Traceability matrix (Golden Thread)

The table below shows the connection from problem found → requirement → solution/feature → test.

Problem / Pain	Requirement	Design / Solution	Feature	Test
Players find long rulebooks difficult to understand	FR-2	Step-by-step learning structure	Interactive Tutorial	Tutorial completion test
Players do not know how rules apply in real situations	FR-3	Example-based learning	Game Situation Practice	Situation response test
Players are unsure whether they understand the rules	FR-4	Understanding assessment	Rule Understanding Check	Answer validation test
Players cannot quickly find a specific rule during gameplay	FR-5	RAG-supported AI Rule Assistant with retrieved sources	AI Rule Assistant	Rule question and source verification test
Players need to clarify an AI answer without repeating the full question	FR-8	Conversation context plus Knowledge Base retrieval	Follow-up Questions	Follow-up context test
Players forget where they stopped learning	FR-6	Persistent progress storage	Learning Progress Tracking	Save and reload test
Learning content needs to be maintained	FR-7	Content management tools	Admin Management	Content update test
AI may lack evidence or become unavailable	NFR-6 / NFR-7	Insufficient-evidence handling, error messages, and response-time measurement	Reliability checks	Unknown-rule and service-failure tests

8. AI usage log

8.1 AI used during SRS preparation

Every use of AI to draft user stories or paragraphs is logged honestly below.

Date	Tool / model	What we ask	What the AI creates	What we adjust / check	What we learn
29 Aug 2026	Claude	Brainstorm feature ideas and draft narratives for initial users for the system to teach board game rules.	Draft user narratives and feature ideas initially, such as lesson plans, step-by-step system instructions, and Q&A.	The team adjusts roles/goals/benefits to align with real problems and adds acceptance criteria that are testable by the users themselves.	AI can help generate good initial ideas, but acceptance criteria need to be developed in-house to be measurable.

8.2 AI used in the product

The proposed product design uses a language model through Ollama and a RAG workflow to retrieve board-game rules from a Knowledge Base. This is distinct from using AI to help prepare the SRS. This section will be updated if the implementation changes.

9. Self-check

- ☒ Every FR is a user story with a benefit — all 8 FRs in 3 end in a "so that" clause tied to a real problem.
- ☒ Every FR has acceptance criteria you could actually test.
- ☒ No FR is vague — no bare "fast", "easy", or "must-have" anywhere in 3.
- ☒ NFRs exist, including security (NFR-4), content accuracy (NFR-5), AI response reliability (NFR-6), and AI response-time measurement (NFR-7).
- ☒ FR-5 describes AI-supported rule questions, Knowledge Base retrieval, and source-aware responses; FR-8 describes follow-up questions and conversational context.
- ☒ Every FR traces back to a stated problem — see 7 Traceability matrix.
- ☒ Scope and out-of-scope are stated — 2.1 and 6.
- ☒ Use-case list matches the FR list — 8 use cases for 8 FRs — 5.
- ☒ AI usage log is honest — 8 distinguishes AI used to prepare the document from AI used by the product.